# MSG No. 74.556 $Im'ma$ [ Type III, orthorhombic ]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m'..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | `[1,2,7,8]` |
| 2 | $\left[0, \frac{1}{2}, 0\right]$ | `[3,4,5,6]` |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[9,10,15,16]` |
| 4 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[11,12,13,14]` |

Table 2: Wyckoff site: `4b`, site symmetry: `2/m'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{2}\right]$ | `[1,2,7,8]` |
| 2 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[3,4,5,6]` |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[9,10,15,16]` |
| 4 | $\left[\frac{1}{2}, 0, 0\right]$ | `[11,12,13,14]` |

Table 3: Wyckoff site: `4c`, site symmetry: `.2'/m.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[1,3,13,15]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[2,4,14,16]` |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[5,7,9,11]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[6,8,10,12]` |

Table 4: Wyckoff site: `4d`, site symmetry: `.2'/m.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[1,3,13,15]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[2,4,14,16]` |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[5,7,9,11]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[6,8,10,12]` |

Table 5: Wyckoff site: `4e`, site symmetry: `m'm2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{4}, z\right]$ | `[1,3,6,8]` |
| 2 | $\left[0, \frac{3}{4}, -z\right]$ | `[2,4,5,7]` |
| 3 | $\left[\frac{1}{2}, \frac{3}{4}, z+\frac{1}{2}\right]$ | `[9,11,14,16]` |
| 4 | $\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$ | `[10,12,13,15]` |

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | `[1,2]` |
| 2 | $\left[x,\ \frac{1}{2},\ 0\right]$ | `[3,4]` |
| 3 | $\left[-x,\ \frac{1}{2},\ 0\right]$ | `[5,6]` |
| 4 | $[-x,\ 0,\ 0]$ | `[7,8]` |
| 5 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[9,10]` |
| 6 | $\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | `[11,12]` |
| 7 | $\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$ | `[13,14]` |
| 8 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[15,16]` |

Table 7: Wyckoff site: `8g`, site symmetry: `.2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$ | `[1,13]` |
| 2 | $\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$ | `[2,14]` |
| 3 | $\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$ | `[3,15]` |
| 4 | $\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{3}{4}\right]$ | `[4,16]` |
| 5 | $\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{3}{4}\right]$ | `[5,9]` |
| 6 | $\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$ | `[6,10]` |
| 7 | $\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$ | `[7,11]` |
| 8 | $\left[\frac{3}{4},\ y,\ \frac{1}{4}\right]$ | `[8,12]` |

Table 8: Wyckoff site: `8h`, site symmetry: `m'..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ z]$ | `[1,8]` |
| 2 | $[0,\ -y,\ -z]$ | `[2,7]` |
| 3 | $\left[0,\ \frac{1}{2}-y,\ z\right]$ | `[3,6]` |
| 4 | $\left[0,\ y+\frac{1}{2},\ -z\right]$ | `[4,5]` |
| 5 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[9,16]` |
| 6 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | `[10,15]` |
| 7 | $\left[\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$ | `[11,14]` |
| 8 | $\left[\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$ | `[12,13]` |

Table 9: Wyckoff site: `8i`, site symmetry: `.m.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{4},\ z\right]$ | `[1,3]` |
| 2 | $\left[x,\ \frac{3}{4},\ -z\right]$ | `[2,4]` |
| 3 | $\left[-x,\ \frac{3}{4},\ -z\right]$ | `[5,7]` |
| 4 | $\left[-x,\ \frac{1}{4},\ z\right]$ | `[6,8]` |
| 5 | $\left[x+\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$ | `[9,11]` |
| 6 | $\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-z\right]$ | `[10,12]` |
| 7 | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}-z\right]$ | `[13,15]` |

*continued ...*

Table 9

| No. | position | mapping |
|---|---|---|
| 8 | $\left[\frac{1}{2}-x,\ \frac{3}{4},\ z+\frac{1}{2}\right]$ | `[14,16]` |

Table 10: Wyckoff site: `16j`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $[x,\ -y,\ -z]$ | `[2]` |
| 3 | $\left[x,\ \frac{1}{2}-y,\ z\right]$ | `[3]` |
| 4 | $\left[x,\ y+\frac{1}{2},\ -z\right]$ | `[4]` |
| 5 | $\left[-x,\ y+\frac{1}{2},\ -z\right]$ | `[5]` |
| 6 | $\left[-x,\ \frac{1}{2}-y,\ z\right]$ | `[6]` |
| 7 | $[-x,\ -y,\ -z]$ | `[7]` |
| 8 | $[-x,\ y,\ z]$ | `[8]` |
| 9 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[9]` |
| 10 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | `[10]` |
| 11 | $\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$ | `[11]` |
| 12 | $\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$ | `[12]` |
| 13 | $\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$ | `[13]` |
| 14 | $\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$ | `[14]` |
| 15 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | `[15]` |
| 16 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[16]` |